

WASP-163b

R-2061

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-163_260415 · created 2026-08-09T01:07:41+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2461145.8933 +/- 0.0074 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.115 +/- 0.014
Transit depth (Rp/Rs)^2	1.33 +/- 0.32 [%]
Orbital Inclination (inc)	83.09 +/- 2.63
Airmass coefficient 1 (a1)	1.055 +/- 0.012
Airmass coefficient 2 (a2)	-0.04 +/- 0.015
Scatter in the residuals of the lightcurve fit is	1.12 %
Best Comparison Star	#1 - [613, 142]
Optimal Aperture	3.26
Optimal Annulus	8.46
Transit Duration (day)	0.075 +/- 0.024

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.115 ± 0.014 vs 0.119 (deviation 0.26σ)
Duration fit vs published	0.075 d vs 0.0942 d (deviation 0.71σ)
Mid-time O–C	7.7 min
Depth SNR	7.3
Residual scatter	1.12 %

Light curve

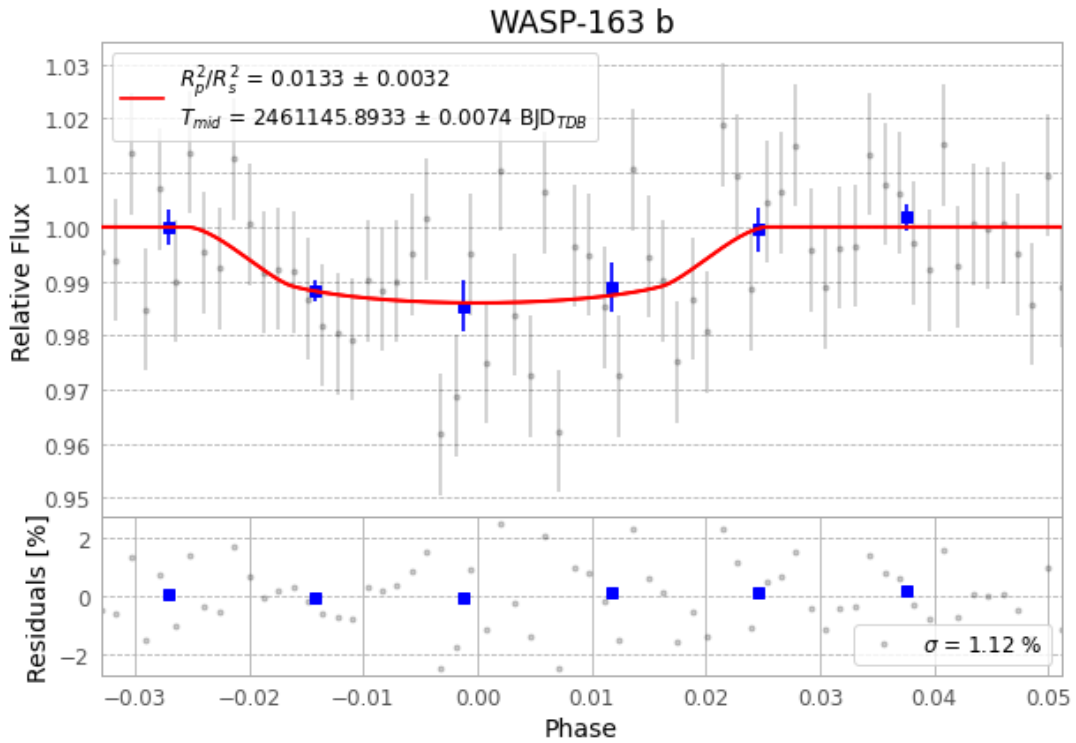


Figure 1 — FinalLightCurve_WASP-163 b_2026-04-15.png (SHA-256 1f5cd190e7b92e45...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [289, 185], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 dfa548a5f402f966...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

66 FITS frames (43 MB), WASP-163260415080315.FITS → WASP-163260415111816.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-163-260415.log (c60d34233010...), FinalLightCurve_WASP-163 b_2026-04-15.png (1f5cd190e7b9...), FinalLightCurve_WASP-163 b_2026-04-15.csv (61ea5e2fa001...)

Compute resources

Wall time 186.5 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-09 01:07Z.